

Algebra II with Trigonometry

Chapter 5.6

Olympiad Solutions (GPT-5.4)

Saved tutoring response with MathJax rendering. Original PDF file:
[4401272_alg-2trig-h-chp-5-6-note-docx.pdf](#)

Solutions

1. A triangle has sides 13, 14, 15. Find the largest angle.

The largest angle is opposite the longest side, so it is the angle opposite side 15.

Let that angle be C . By the Law of Cosines,

$$15^2 = 13^2 + 14^2 - 2(13)(14) \cos C.$$

Thus

$$225 = 169 + 196 - 364 \cos C,$$

$$225 = 365 - 364 \cos C,$$

$$364 \cos C = 140,$$

$$\cos C = \frac{140}{364} = \frac{5}{13}.$$

Hence

$$C = \cos^{-1}\left(\frac{5}{13}\right).$$

So the largest angle is

$$\cos^{-1}\left(\frac{5}{13}\right) \approx \boxed{67.4^\circ}.$$

2. Two sides of a triangle are 20 and 21, and the included angle is 60° . Find the third side, then the area.

Let the third side be c . By the Law of Cosines,

$$c^2 = 20^2 + 21^2 - 2(20)(21) \cos 60^\circ.$$

Since $\cos 60^\circ = \frac{1}{2}$,

$$c^2 = 400 + 441 - 840 \cdot \frac{1}{2} = 841 - 420 = 421.$$

So

$$\boxed{c = \sqrt{421}}.$$

For the area, use

$$[\triangle] = \frac{1}{2}ab \sin C.$$

Thus

$$[\triangle] = \frac{1}{2}(20)(21) \sin 60^\circ = 210 \cdot \frac{\sqrt{3}}{2} = 105\sqrt{3}.$$

Therefore the answers are

$$\boxed{\sqrt{421}} \quad \text{and} \quad \boxed{105\sqrt{3}}.$$

3. A triangle has sides 25, 25, 30. Find its area without using a calculator.

This is an isosceles triangle with base 30. Drop an altitude from the apex to the base; it bisects the base into two segments of length 15.

Then the height h satisfies

$$h^2 + 15^2 = 25^2.$$

So

$$h^2 = 625 - 225 = 400,$$

$$h = 20.$$

Hence the area is

$$\frac{1}{2} \cdot 30 \cdot 20 = 300.$$

Therefore the area is

$$\boxed{300}.$$

4. A hiker walks 12 km due east, then $5\sqrt{2}$ km on a bearing $N45^\circ E$. How far is the hiker from the starting point?

Take east as the x -direction and north as the y -direction.

First displacement: 12 km due east gives

$$(12, 0).$$

Second displacement: $5\sqrt{2}$ km on a bearing $N45^\circ E$ means 45° east of north, so its east and north components are equal:

$$5\sqrt{2} \cdot \cos 45^\circ = 5, \quad 5\sqrt{2} \cdot \sin 45^\circ = 5.$$

Thus the second displacement is

$$(5, 5).$$

Total displacement:

$$(12, 0) + (5, 5) = (17, 5).$$

Distance from the start:

$$\sqrt{17^2 + 5^2} = \sqrt{289 + 25} = \sqrt{314}.$$

So the hiker is

$$\boxed{\sqrt{314} \text{ km}}$$

from the starting point.

5. Point B is 100 miles from point A at bearing $N60^\circ E$. A pilot mistakenly flies due east from A for 50 miles. How far is the pilot from B ?

Place A at the origin, with east as x -axis and north as y -axis.

Bearing $N60^\circ E$ means 60° east of north, equivalently 30° north of east.

So the coordinates of B are

$$(100 \cos 30^\circ, 100 \sin 30^\circ) = (50\sqrt{3}, 50).$$

The pilot flies due east 50 miles, so the pilot's position is

$$P = (50, 0).$$

Hence

$$PB^2 = (50\sqrt{3} - 50)^2 + (50 - 0)^2.$$

Factor out 50^2 :

$$PB^2 = 50^2 [(\sqrt{3} - 1)^2 + 1].$$

Now

$$(\sqrt{3} - 1)^2 = 3 - 2\sqrt{3} + 1 = 4 - 2\sqrt{3},$$

so

$$PB^2 = 2500(5 - 2\sqrt{3}).$$

Therefore

$$PB = 50\sqrt{5 - 2\sqrt{3}}.$$

So the pilot is

$$\boxed{50\sqrt{5 - 2\sqrt{3}} \text{ miles}}$$

from B .

6. A triangle has sides 29, 29, 42. Find the exact value of $\cos \theta$, where θ is one of the base angles.

The triangle is isosceles with equal sides 29, 29 and base 42. A base angle is the angle between sides 29 and 42, opposite the other side 29.

Apply the Law of Cosines:

$$29^2 = 29^2 + 42^2 - 2(29)(42) \cos \theta.$$

Cancel 29^2 from both sides:

$$0 = 42^2 - 2(29)(42) \cos \theta.$$

So

$$2(29)(42) \cos \theta = 42^2.$$

Divide by 42:

$$2(29) \cos \theta = 42.$$

Thus

$$\cos \theta = \frac{42}{58} = \frac{21}{29}.$$

Therefore

$$\cos \theta = \frac{21}{29}.$$